

EXPERIMENT 03

Java Program to Display Farm Animal Details Using Classes and Objects

■ Basic Information

Experiment No: 03

Date: 11/09/2026

Title: Java Program to Display Farm Animal Details Using Classes and Objects

■ Objective / Aim

To develop a Java program using a class and objects to store and display the details of different farm animals such as their name, place, food, and sound.

■ System Requirements

Hardware

- Standard PC/Laptop
- Minimum 4 GB RAM
- Minimum 500 MB free storage

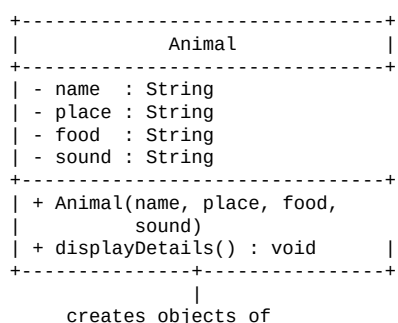
Software

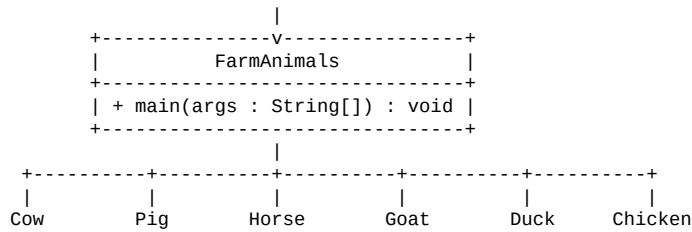
- Windows / Linux / macOS
- Java Development Kit (JDK)
- Java IDE such as Eclipse, IntelliJ IDEA, NetBeans, or VS Code
- Command Prompt / Terminal

■ Algorithm / Logic Description

1. Start the program.
2. Create a class named `Animal`.
3. Declare the data members `name`, `place`, `food`, and `sound`.
4. Create a constructor to initialize the animal details.
5. Create a `displayDetails()` method to display the details of an animal.
6. Create the main class named `FarmAnimals`.
7. Create objects for `Cow`, `Pig`, `Horse`, `Goat`, `Duck`, and `Chicken`.
8. Pass the `name`, `place`, `food`, and `sound` values to each object through the constructor.
9. Call `displayDetails()` for each animal object.
10. Display the details of all farm animals.
11. Stop the program.

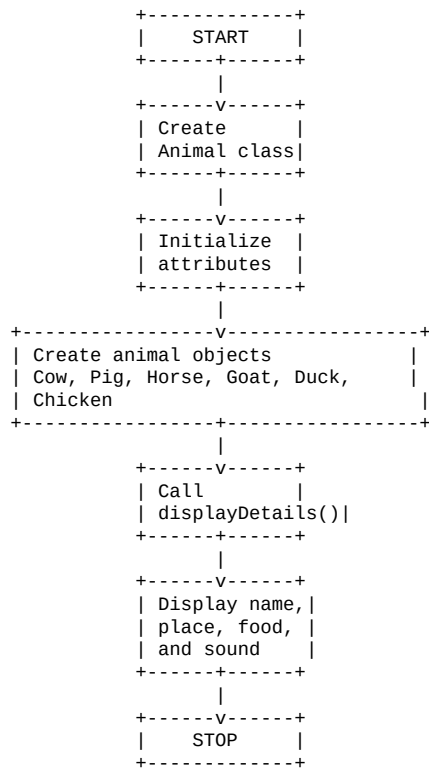
■ UML Class Diagram





Relationship: The FarmAnimals class contains the main() method and creates multiple objects of the Animal class. Each object stores the details of one farm animal.

■ Program Flow Diagram



■ Source Code

File name: FarmAnimals.java

```
package recordprog;

// Base class
class Animal {

    String name;
    String place;
    String food;
    String sound;

    Animal(String name, String place, String food, String sound) {
        this.name = name;
        this.place = place;
        this.food = food;
        this.sound = sound;
    }

    void displayDetails() {
        System.out.println("The " + name + " lives in the " + place +
            ", eats " + food +
            ", and makes the sound \"" + sound + "\".");
    }
}

public class FarmAnimals {

    public static void main(String[] args) {

        Animal cow = new Animal(
            "Cow", "cowshed", "grass & hay", "Moo Moo");

        Animal pig = new Animal(
            "Pig", "pigsty", "vegetables & grains", "Oink Oink");

        Animal horse = new Animal(
            "Horse", "stable", "hay & oats", "Neigh Neigh");

        Animal goat = new Animal(
            "Goat", "goat pen", "leaves & grass", "Baa Baa");

        Animal duck = new Animal(
            "Duck", "pond", "insects & grains", "Quack Quack");

        Animal chicken = new Animal(
            "Chicken", "henhouse", "grains & seeds", "Cluck Cluck");

        cow.displayDetails();
        pig.displayDetails();
        horse.displayDetails();
        goat.displayDetails();
        duck.displayDetails();
        chicken.displayDetails();
    }
}
```

■ Compilation & Execution Commands

Since the program contains package `recordprog`;, create a folder named `recordprog` and save the file inside it as:

```
recordprog/FarmAnimals.java
```

Compile:

```
javac recordprog/FarmAnimals.java
```

Run:

```
java recordprog.FarmAnimals
```

■■ Sample Output

```
The Cow lives in the cowshed, eats grass & hay, and makes the sound "Moo Moo".
The Pig lives in the pigsty, eats vegetables & grains, and makes the sound "Oink Oink".
The Horse lives in the stable, eats hay & oats, and makes the sound "Neigh Neigh".
The Goat lives in the goat pen, eats leaves & grass, and makes the sound "Baa Baa".
The Duck lives in the pond, eats insects & grains, and makes the sound "Quack Quack".
The Chicken lives in the henhouse, eats grains & seeds, and makes the sound "Cluck Cluck".
```

■ Result

The Java program to display farm animal details was successfully implemented using **classes, objects, constructors, data members, and methods**. The program successfully creates objects for different farm animals and displays their name, place, food, and sound.